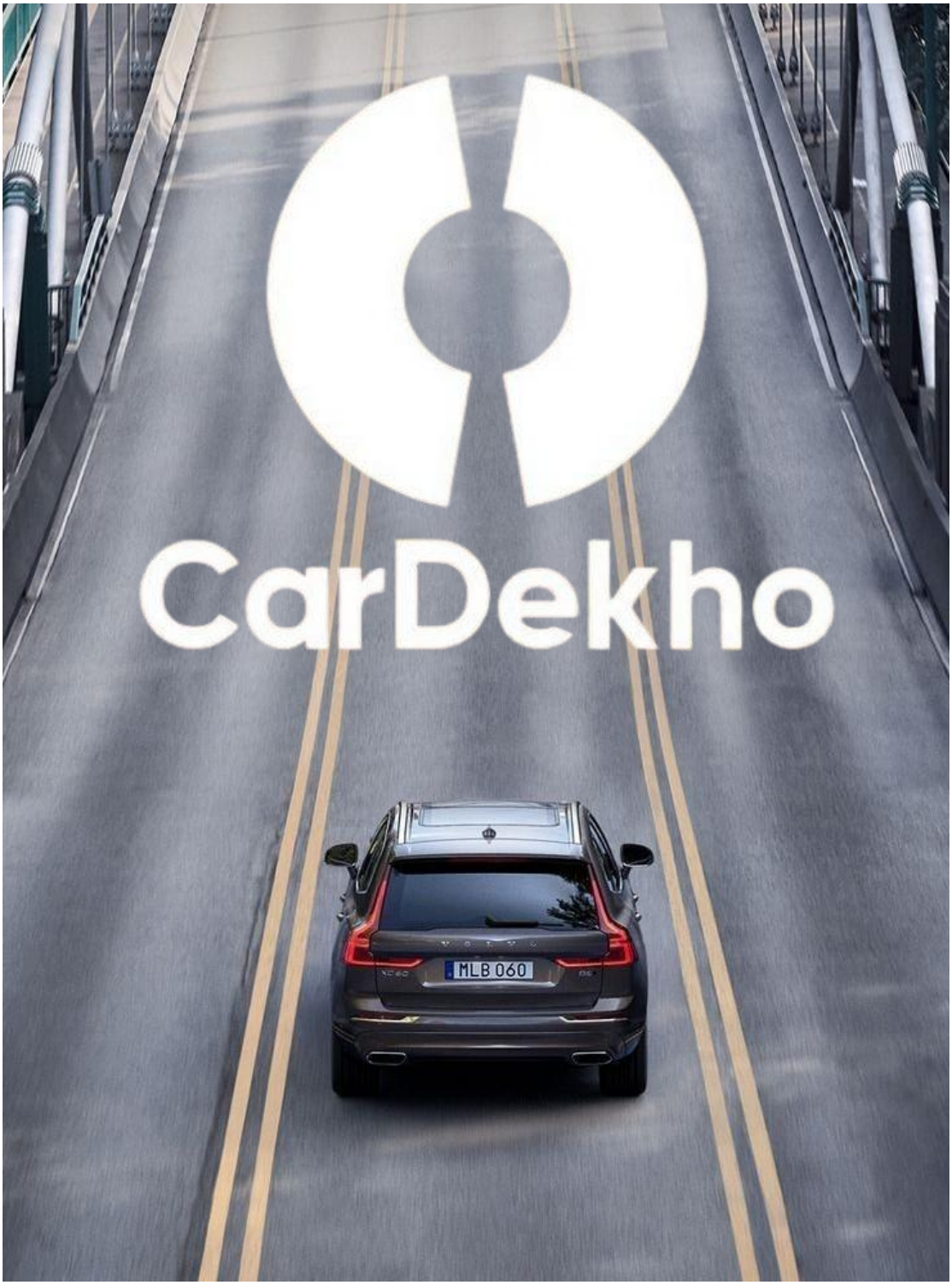
CAR DEKHO DATASET ANALYSIS



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INTRODUCTION

The used-car market contains vehicles with different ages, fuel types, ownership histories, specifications, and usage levels. Understanding these characteristics is important for identifying patterns in vehicle pricing and market composition.

The objective of this project is to analyze the Car Dekho used-car dataset and identify meaningful patterns in:

- Vehicle characteristics
- Market composition
- Selling prices
- Vehicle age
- Usage
- Engine and power specifications
- Mileage
- Fuel type
- Transmission
- Ownership
- Seller type

The analysis was performed using **MySQL**, followed by exploratory analysis of the cleaned and transformed dataset.

Dataset Overview

The dataset contains information about used vehicles, including:

- name
- year

- selling_price
- km_driven
- fuel
- seller_type
- transmission
- owner
- mileage
- engine
- max_power
- torque
- seats

The initial stage of the project focused on understanding the available columns and checking the quality and structure of the data.

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1. Data Cleaning

1. Missing Value Analysis

A missing-value audit was performed across the major columns of the dataset.

For text-based fields, both NULL values and empty strings were checked, while numerical fields were checked for NULL values. This helped identify incomplete records before performing further analysis.

The analysis specifically identified missing values in the **torque** field. Records with missing torque values were subsequently removed to improve consistency for further analysis.

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2. Duplicate Analysis

Duplicate records were investigated using a combination of vehicle attributes such as:

- Vehicle name
- Year
- Selling price
- Kilometers driven
- Fuel
- Seller type
- Transmission
- Owner
- Mileage
- Engine
- Maximum power

- Torque
- Seats

Records occurring more than once were identified using GROUP BY and HAVING COUNT(*) > 1.

A cleaned table was then created using distinct records, and the cleaned data was retained for subsequent analysis.

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5. Data Transformation (data into proper format)

Several vehicle specifications contained both numerical values and units.

For example:

- Mileage contained numerical values along with mileage units.
- Engine contained engine capacity and CC.
- Maximum power contained numerical values and bhp.
- Torque contained numerical values and Nm.

Instead of modifying the original fields and removing their units, a separate **car_dekho_extra** table was created to store structured versions of these attributes.

The numerical values were extracted using SQL string and conversion functions such as SUBSTRING_INDEX, REPLACE, and CAST.

This approach preserved the original information while making the numerical specifications suitable for analysis.

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2. Feature Engineering

To support deeper analysis, several new features were created in the car_dekho_extra table.

2.1 Car Age

Car age was calculated using:

2026 - year

This converts the manufacturing year into a more meaningful analytical variable: **vehicle age**.

2.2 Engine Category

Engine capacity was classified into:

- Small
- Medium
- Large
- Very Large

This allows vehicles to be compared based on engine-size groups rather than individual engine values.

2.3 Power Category

Maximum power was categorized into:

- Low
- Medium
- High
- Very High

2.4 Torque Category

Torque was classified into:

- Low
- Medium
- High
- Very High

2.5 Mileage Category

Mileage was grouped into:

- Low
- Medium
- High

---

3 . Exploratory Data Analysis

The exploratory analysis was divided into three levels:

Univariate Analysis

Analysis of individual variables and their distributions.

Bivariate Analysis

Analysis of relationships between two variables, particularly vehicle characteristics and selling price.

Multivariate Analysis

Analysis of the combined effect of multiple vehicle characteristics on selling price.

The SQL analysis contains queries covering these three levels.

QUESTIONS AND QUERIES

A. Univariate Analysis

Analysis of one variable at a time.

1. Which fuel types dominate the used-car market?

```
SELECT
    fuel,
    COUNT(*) AS number_of_cars
FROM car_dekho
GROUP BY fuel
ORDER BY number_of_cars DESC;
```

2. How is the used-car market distributed by ownership?

```
SELECT
    owner,
    COUNT(*) AS number_of_cars
FROM car_dekho
GROUP BY owner
ORDER BY number_of_cars DESC;
```

3. Which seller type dominates the dataset?

```
SELECT
    seller_type,
    COUNT(*) AS number_of_cars
FROM car_dekho
GROUP BY seller_type
ORDER BY number_of_cars DESC;
```

4. What is the distribution of selling prices?

```
SELECT
    CASE
        WHEN selling_price < 100000 THEN '0 - 99,999'
        WHEN selling_price < 200000 THEN '100,000 - 199,999'
        WHEN selling_price < 300000 THEN '200,000 - 299,999'
        WHEN selling_price < 400000 THEN '300,000 - 399,999'
        WHEN selling_price < 500000 THEN '400,000 - 499,999'
        WHEN selling_price < 600000 THEN '500,000 - 599,999'
        WHEN selling_price < 700000 THEN '600,000 - 699,999'
        WHEN selling_price < 800000 THEN '700,000 - 799,999'
```

```
        WHEN selling_price < 900000 THEN '800,000 - 899,999'
        ELSE '900,000+'
    END AS price_range,
    COUNT(*) AS number_of_cars
FROM car_dekho
GROUP BY price_range
ORDER BY MIN(selling_price);
```

5. Which manufacturing years have the highest number of cars?

```
SELECT
    year,
    COUNT(*) AS number_of_cars
FROM car_dekho
GROUP BY year
ORDER BY year;
```

B. Bivariate Analysis

1. Which fuel type has the highest average selling price?

```
SELECT
    fuel,
    AVG(selling_price) AS average_selling_price
FROM car_dekho
GROUP BY fuel
ORDER BY average_selling_price DESC;
```

2. Does car age affect selling price?

```
SELECT
    e.car_age,
    AVG(c.selling_price) AS average_selling_price
FROM car_dekho_extra e
JOIN car_dekho c
    ON e.id = c.id
GROUP BY e.car_age
ORDER BY e.car_age;
```

3. Does transmission type affect average selling price?

```
SELECT
    transmission,
    AVG(selling_price) AS average_selling_price
FROM car_dekho
GROUP BY transmission
ORDER BY average_selling_price DESC;
```

4. Does engine capacity affect average selling price?

```
SELECT
    e.engine,
    AVG(c.selling_price) AS average_selling_price
FROM car_dekho_extra e
JOIN car_dekho c
    ON e.id = c.id
GROUP BY e.engine
```

```
ORDER BY e.engine;
```

5. Does maximum power affect average selling price?

```
SELECT
    e.max_power,
    AVG(c.selling_price) AS average_selling_price
FROM car_dekho_extra e
JOIN car_dekho c
    ON e.id = c.id
GROUP BY e.max_power
ORDER BY e.max_power;
```

C. Multivariate Analysis

1. How do car age and kilometres driven together relate to selling price?

```
SELECT
    e.car_age,
    AVG(c.km_driven) AS average_km_driven,
    AVG(c.selling_price) AS average_selling_price
FROM car_dekho_extra e
JOIN car_dekho c
    ON e.id = c.id
GROUP BY e.car_age
ORDER BY e.car_age;
```

2. How are engine capacity and maximum power together related to selling price?

```
SELECT
    e.engine,
    e.max_power,
    c.selling_price
FROM car_dekho_extra e
JOIN car_dekho c
    ON e.id = c.id
ORDER BY e.engine;
```

3. How does average selling price vary across fuel types and transmission types?

```
SELECT
    c.fuel,
    c.transmission,
    AVG(c.selling_price) AS average_selling_price
FROM car_dekho c
GROUP BY c.fuel, c.transmission
ORDER BY c.fuel, c.transmission;
```

4. How do fuel type, car age, and selling price relate to each other?

```
SELECT
    c.fuel,
    e.car_age,
    AVG(c.selling_price) AS average_selling_price
```

```
FROM car_dekho c
JOIN car_dekho_extra e
    ON c.id = e.id
GROUP BY c.fuel, e.car_age
ORDER BY e.car_age;
```

5. How do engine category, fuel type, and selling price relate?

```
SELECT
    e.engine_category,
    c.fuel,
    AVG(c.selling_price) AS average_selling_price
FROM car_dekho_extra e
JOIN car_dekho c
    ON e.id = c.id
GROUP BY e.engine_category, c.fuel
ORDER BY e.engine_category, c.fuel;
```

Market Overview Findings

1. Overall Dataset

The analysis contains approximately **6.73K vehicles**.

The key overall indicators are:

Metric	Value
Total Cars	6.73K
Average Selling Price	528.96K
Average KM Driven	73.21K
Approx. Average Car Age	12.36 years
Average Mileage	19.47

These values provide an overall picture of the used-car dataset before examining individual characteristics.

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2. Cars by Fuel Type

The visualization shows that **Diesel and Petrol vehicles dominate the dataset**.

Approximately:

- Diesel: **3.7K**
- Petrol: **3.0K**
- CNG: **0.1K**
- LPG: very small representation
- Electric: very small representation

Finding

The used-car dataset is heavily concentrated around conventional fuel vehicles, particularly Diesel and Petrol. Alternative-fuel vehicles such as CNG, LPG, and Electric represent only a small portion of the available vehicles.

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3. Ownership Analysis

The ownership distribution shows that **First Owner vehicles are the dominant category**.

The number of vehicles decreases considerably as the ownership count increases:

First Owner → Second Owner → Third Owner → Fourth & Above Owner

Test Drive Cars represent only a very small portion of the dataset.

Finding

First-owner vehicles form the largest segment of the used-car market represented in the dataset, while vehicles with multiple previous owners are progressively less common.

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4. Manufacturing Year Analysis

The year-wise visualization shows that the number of vehicles varies significantly across manufacturing years.

The number of vehicles increases substantially for newer model years, with the highest observed count around **2017**, reaching approximately **802 vehicles** in the visualization.

Finding

The dataset contains a substantially larger number of vehicles from relatively recent manufacturing years compared with the oldest vehicles, indicating a stronger representation of newer used cars.

---

5. Transmission Analysis

The transmission distribution shows a strong dominance of **Manual vehicles**.

Approximately:

- Manual: **91.36%**
- Automatic: **8.64%**

Finding

Manual transmission vehicles overwhelmingly dominate the dataset, accounting for more than nine out of every ten vehicles, while automatic vehicles represent a relatively small segment.

---

6. Seller Type Analysis

The seller-type visualization shows that **Individual sellers dominate the dataset**, followed by Dealers and Trustmark Dealers.

Finding

The used-car listings are predominantly associated with individual sellers, while dealer and Trustmark Dealer listings represent smaller portions of the market.

---

7. Selling Price Analysis

The selling-price distribution is strongly concentrated toward the lower price range.

The visualization shows a high concentration of vehicles at relatively lower prices, followed by a long tail extending toward higher-priced vehicles.

The reported price range is approximately:

- **Minimum:** 30K
- **Maximum:** 10M
- **Average:** 528.96K

Finding

The used-car market exhibits a strongly right-skewed price distribution. Most vehicles are concentrated in the lower price range, while a relatively small number of high-value vehicles extend the upper end of the price distribution.

This indicates that the average selling price should be interpreted carefully because a small number of expensive vehicles can influence the overall average.

---

8. Price Analysis by Fuel Type

The average selling price differs substantially across fuel types.

The visualization shows approximately:

Fuel Type	Average Selling Price
Electric	2.65M
Diesel	0.65M
Petrol	0.39M
CNG	0.32M
LPG	0.21M

Finding

Electric vehicles have the highest average selling price in the analyzed dataset, followed by Diesel vehicles. Petrol, CNG, and LPG vehicles have considerably lower average selling prices.

However, the Electric category has very few vehicles in the dataset. Therefore, its high average price should be interpreted in the context of its small sample size.

---

9. Price Analysis by Car Age

The visualization shows a clear downward trend in average selling price as **car age increases**.

The youngest vehicles have the highest average prices, while older vehicles show substantially lower average prices.

For example, the visualization shows an average price of approximately **1.9M** at a very low car age, followed by a progressive decline as age increases.

Finding

Vehicle age shows a strong inverse relationship with average selling price. As cars become older, their average selling price generally decreases.

This is one of the clearest pricing patterns observed in the analysis.

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10. Price Analysis by Transmission

The visualization shows a substantial difference between automatic and manual vehicles.

Approximate average selling prices:

- **Automatic:** 1.34M

- **Manual:** 0.45M

Finding

Automatic vehicles have a substantially higher average selling price than manual vehicles in the analyzed dataset.

The difference suggests that transmission type is an important factor associated with used-car prices. However, this relationship may also be influenced by other factors such as vehicle age, brand, engine capacity, and fuel type.

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11. Price Analysis by Engine

The engine-versus-price visualization shows that vehicles with larger engine capacities can achieve higher selling prices.

However, the relationship is not perfectly linear. Vehicles with similar engine capacities can have considerably different selling prices.

Finding

Engine capacity is associated with selling price, but engine size alone does not fully explain vehicle pricing. Other vehicle characteristics also contribute to the observed price differences.

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12. Price Analysis by Ownership

The average selling-price visualization shows noticeable differences across ownership categories.

Test Drive Cars display a particularly high average selling price compared with other ownership categories, while First Owner, Second Owner, Third Owner, and Fourth & Above Owner vehicles have substantially lower averages.

Important interpretation

Because the number of Test Drive Cars is very small compared with First Owner and Second Owner vehicles, its unusually high average should be interpreted cautiously.

Finding

Ownership category is associated with differences in average selling price, but categories with very small numbers of vehicles may produce unstable or unusually high average values.

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13. Engine Category Analysis

The engine-category visualization shows that:

- **Medium** engine vehicles are the largest category.
- **Small** engine vehicles are also highly represented.
- **Very Large** engine vehicles form a smaller segment.
- **Large** engine vehicles have the smallest representation among the displayed categories.

Finding

The used-car dataset is primarily composed of Small and Medium engine vehicles, while Large and Very Large engine vehicles represent comparatively smaller segments.

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14. Power Category Analysis

The power-category distribution shows that **Medium and Low power vehicles dominate the dataset**.

High-power vehicles are considerably fewer, while Very High-power vehicles form the smallest category.

Finding

Most vehicles in the dataset fall into the Low and Medium power categories, indicating that high-performance vehicles represent only a relatively small segment of the used-car market.

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15. Mileage Analysis

The mileage-category visualization shows that **High mileage vehicles form the largest category**, followed by Medium and Low mileage vehicles.

The fuel-wise average mileage visualization shows:

Fuel Type	Average Mileage
Electric	32.5
CNG	24.2
Diesel	19.7
Petrol	19.1
LPG	18.6

Finding

Electric vehicles show the highest average mileage in the analyzed dataset, followed by CNG vehicles. Diesel and Petrol vehicles show similar average mileage values, while LPG has the lowest average among the displayed fuel types.

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16. Price by Car Age and Fuel Type

The combined analysis of car age and fuel type shows that **average selling price generally decreases as vehicle age increases**.

The visualization also demonstrates that the price pattern differs between fuel types, with some fuel categories showing substantially higher prices at younger vehicle ages.

Finding

Car age remains an important factor in pricing across fuel categories. Although the magnitude of the decline differs between fuel types, older vehicles generally command lower average selling prices.

Key Findings

Based on the overall analysis, the major findings are:

1. Market Composition

The dataset is dominated by **Diesel and Petrol vehicles**.

2. Ownership

First-owner vehicles represent the largest ownership category.

3. Transmission

Manual transmission vehicles account for approximately 91.36% of the dataset.

4. Seller Type

Individual sellers dominate the seller-type distribution.

5. Price Distribution

Selling prices are **strongly concentrated toward the lower end**, with a relatively small number of expensive vehicles.

6. Vehicle Age

Average selling price generally **decreases as car age increases**.

7. Fuel and Price

Electric vehicles have the **highest average selling price** in the visualization, although they represent a very small portion of the dataset.

8. Transmission and Price

Automatic vehicles have a **substantially higher average selling price** than manual vehicles.

9. Engine

Small and Medium engine vehicles form the largest portions of the dataset.

10. Power

Low and Medium power vehicles dominate the dataset.

11. Mileage

Electric vehicles show the highest average mileage, followed by CNG.

12. Engine and Price

Larger engine capacity is associated with higher prices in some cases, but engine size alone does not determine selling price.

Business Insights

The analysis provides several useful insights into the used-car market.

For Buyers

Buyers looking for lower-priced vehicles are likely to find more options among older, manual, Petrol and Diesel vehicles.

Buyers interested in newer or automatic vehicles should generally expect higher prices based on the observed dataset patterns.

For Sellers

Newer vehicles usually get better prices. Highlighting features like transmission, engine, and mileage can also help attract buyers and improve the vehicle’s value.

For Market Analysis

The strong price skew suggests that **median price and segmented analysis** can be useful alongside average price when evaluating the used-car market.

Conclusion

The Car Dekho used-car dataset provides a comprehensive view of vehicle characteristics and pricing patterns in the used-car market. Through systematic data cleaning, duplicate removal, data transformation, feature engineering, and exploratory data analysis, the dataset was prepared for detailed investigation.

The analysis shows that the market is predominantly composed of Diesel and Petrol vehicles, first-owner vehicles, manual transmissions, and low-to-medium power and engine categories. Selling prices are strongly concentrated toward the lower range, while a small number of high-value vehicles create a long upper tail.

Vehicle age shows a clear negative relationship with selling price, with newer vehicles generally commanding higher prices. Automatic vehicles have considerably higher average selling prices than manual vehicles, while engine capacity also shows an association with price. Electric vehicles demonstrate the highest average selling price and mileage in the analyzed visualizations, although their representation in the dataset is very small.

Overall, the analysis demonstrates that used-car pricing is influenced by multiple factors rather than a single vehicle characteristic. Car age, fuel type, transmission, engine capacity, mileage, ownership, and other vehicle specifications collectively contribute to the observed differences in selling prices.

26. Project Workflow

You can summarize the entire project in your report as:

Raw Car Dekho Dataset
↓
Data Understanding
↓
Data Type & Missing Value Analysis
↓
Duplicate Identification & Removal
↓
Data Transformation
↓
Feature Engineering
↓
Univariate Analysis
↓
Bivariate Analysis
↓
Multivariate Analysis
↓
Key Findings & Business Insights
↓
Conclusion

Powerbi Report Link:

https://app.powerbi.com/groups/me/reports/cacfeffd-087e-4ea5-b154-badbef1e6cb2/d40ecd87cbbc03212a1e?experience=power-bi

Visualization pdf Link :
https://drive.google.com/file/d/1pVstMBBmma_ZnAf1Q09C_ihtEEfBK0e7/view?usp=sharing